FILTER DESIGN TEST CASE

BAW (Bulk Acoustic Wave) - Band 41 (2500 MHz TDD) for GPS/GNSS Anti-Jamming Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Author:	Dr. Takeshi Yamamoto
Reviewer:	Dr. Anya Krishnamurthy
Design Center:	Helsinki 5G Lab
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1. OBJECTIVE

This document describes the design, simulation, and characterization of a BAW (Bulk Acoustic Wave) filter targeting Band 41 (2500 MHz TDD) for GPS/GNSS Anti-Jamming Filter. The filter is designed on 128° YX $LiNbO_3$ substrate with a target center frequency of 712.1 MHz and bandwidth of 91.5 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the GPS/GNSS Anti-Jamming Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: BAW (Bulk Acoustic Wave)
- Frequency Band: Band 41 (2500 MHz TDD)
- Application: GPS/GNSS Anti-Jamming Filter
- Substrate: 128° YX $LiNbO_3$
- Package: Flip-Chip BGA
- Design Tool: Cadence AWR Microwave Office
- Design Center: Helsinki 5G Lab

This specification applies to all variants of the BAW (Bulk Acoustic Wave) design targeting Band 41 (2500 MHz TDD). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 712.1 MHz
Bandwidth (3 dB): 91.5 MHz
Insertion Loss Target: <= 2.2 dB
Return Loss Target: >= 11.0 dB
Out-of-Band Rejection: >= 23.0 dB
Isolation (Duplexer): >= 44.0 dB
Q Factor: 12862
Temperature Coefficient:-29.5 ppm/°C
Die Size: 1.5 x 1.7 mm
Wafer Size: 6-inch
Number of Resonators: 9
Electrode Material: Pt/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Cadence AWR Microwave Office and load the BAW (Bulk Acoustic Wave) template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 712.1 MHz, BW 91.5 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Rohde & Schwarz ZNB40.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 2136 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.2 dB
- Return loss in passband shall be >= 11.0 dB
- Out-of-band rejection at +/- 137 MHz offset >= 23.0 dB
- Group delay variation across passband shall be <= 7 ns
- Temperature drift over -40°C to +85°C shall be <= 2.6 MHz
- Power handling shall be >= +31 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2024-02-21
Devices Tested: 67
Insertion Loss (meas): 1.95 dB
Return Loss (meas): 13.4 dB
Rejection (meas): 25.9 dB
Isolation (meas): 49.1 dB
Q Factor (meas): 12862
Group Delay Variation: 6.9 ns
Power Handling: +28 dBm
Die Yield: 93.6%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

a) The BAW (Bulk Acoustic Wave) design demonstrated below-target performance for Band 41 (2500 MHz TDD).
b) Insertion loss met target specification.
c) Rejection at near-band offset requires additional resonator tuning.
d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

Author: Dr. Takeshi Yamamoto Date: 2024-02-21
Reviewer: Dr. Anya Krishnamurthy Date: 2024-02-27
Approver: Dr. Yuko Ishida Date: 2024-02-25

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